

$$7. (a) X(s) = \frac{1}{s^2 + 5s + 4} = \frac{1}{(s+1)(s+4)} = \frac{1}{3} \cdot \frac{1}{(s+1)(s+4)}$$

$$\Rightarrow x(t) = \mathcal{L}^{-1}\{X(s)\} = \frac{1}{3}(e^{-t} - e^{-4t})$$

$$\Rightarrow x(kT) = \frac{1}{3}(e^{-kT} - e^{-4kT})$$

$$\begin{aligned} \Rightarrow X(z) &= \frac{(e^{-T} - e^{-4T})z^{-1}}{3(1 - e^{-T}z^{-1})(1 - e^{-4T}z^{-1})} \\ &= \frac{0.4712z^{-1}}{3(1 - 0.6065z^{-1})(1 - 0.1353z^{-1})} \end{aligned}$$

$$(b) x(k+2) + (\alpha + 0.5)x(k+1) + 0.5\alpha x(k) = \delta(k)$$

$$\begin{aligned} \Rightarrow z^2 X(z) - z^2 x(0) - z x(1) + (\alpha + 0.5)z X(z) - (\alpha + 0.5)z x(0) \\ + 0.5\alpha X(z) = 1 \end{aligned}$$

$$\Rightarrow [z^2 + (\alpha + 0.5)z + 0.5\alpha] X(z) - z^2 x(0) - z x(1) - (\alpha + 0.5)z x(0) = 1$$

$$k = -2: x(0) = \delta(-2) = 0$$

$$k = -1: x(1) + (\alpha + 0.5)x(0) = \delta(-1) = 0, x(1) = 0$$

$$k = 0: x(2) + (\alpha + 0.5)x(1) + 0.5\alpha x(0) = \delta(0) = 1, x(2) = 1$$

$$\Rightarrow [z^2 + (\alpha + 0.5)z + 0.5\alpha] X(z) = 1, X(z) = \frac{1}{z^2 + (\alpha + 0.5)z + 0.5\alpha}$$

$$(A+B)z + 0.5A + \alpha B = 1 \quad = \frac{1}{(z+\alpha)(z+0.5)}$$

$$\therefore A+B=0, A=-B$$

$$0.5A + \alpha B = 1, (\alpha - 0.5)B = 1$$

$$B = \frac{1}{\alpha - 0.5}, A = \frac{-1}{\alpha - 0.5}$$

$$= \frac{A}{z+\alpha} + \frac{B}{z+0.5}$$

$$\Rightarrow X(z) = \frac{1}{0.5 - \alpha} \times \frac{1}{z + \alpha} - \frac{1}{0.5 - \alpha} \times \frac{1}{z + 0.5}$$

$$\Rightarrow x(k) = \frac{(-\alpha)^{k-1}}{0.5 - \alpha} - \frac{(-0.5)^{k-1}}{0.5 - \alpha} = \frac{(0.5)^{k-1} - (-\alpha)^{k-1}}{\alpha - 0.5}$$

$$(c) x(\infty) = \lim_{k \rightarrow \infty} x(k) = \lim_{z \rightarrow 1} (1 - z^{-1}) X(z)$$

$$= \lim_{z \rightarrow 1} \frac{z^{-2}}{(1 - z^{-1})(1 + \alpha z^{-1})(1 + 0.5z^{-1})}$$

$$= \lim_{z \rightarrow 1} \frac{z^{-2}}{(1 - z^{-1})(1 + \alpha z^{-1})(1 + 0.5z^{-1})}$$

$$\begin{aligned} \text{4. (a)} \quad E(s) &= R(s) - H(s)C(s), \\ C(s) &= E^*(s) G_1(s) G_2(s), \quad C^*(s) = E^*(s) [G_1 G_2(s)]^* \\ \Rightarrow E^*(s) &= R^*(s) - H^*(s) [G_1 G_2(s)]^* E^*(s) \\ \Rightarrow E^*(s) &= \frac{R^*(s)}{1 + H^*(s) [G_1 G_2(s)]^*} \end{aligned}$$

$\Rightarrow$ pulse transfer function:

$$\begin{aligned} \frac{C^*(s)}{R^*(s)} &= \frac{[G_1 G_2(s)]^*}{1 + H^*(s) [G_1 G_2(s)]^*} \\ \frac{C(z)}{R(z)} &= \frac{G_1 G_2(z)}{1 + H(z) G_1 G_2(z)} \end{aligned}$$

$$\begin{aligned} \text{(b)} \quad R(z) &= 1, \quad G_1 G_2(z) = z [G_1(s) G_2(s)] \\ &= (1-z^{-1}) z \left[\frac{G_2(s)}{s} \right] = (1-z^{-1}) z \left[\frac{\beta}{s^2(s+1)} \right] \\ &= \beta (1-z^{-1}) \frac{[(1-1+e^{-T}) + (1-e^{-T} - T e^{-T})z^{-1}] z^{-1}}{(1-z^{-1})^2 (1-e^{-T}z^{-1})} \\ &= \frac{\beta [0.00484 + 0.00468z^{-1}] z^{-1}}{(1-z^{-1})(1-0.9048z^{-1})} \end{aligned}$$

$$1 + G_1 G_2(z) = \frac{1 - 0.9048z^{-1} + 0.9048z^{-2} + \beta 0.00484z^{-1} + \beta 0.00468z^{-2}}{(1-z^{-1})(1-0.9048z^{-1})}$$

$$\Rightarrow C(z) = \frac{G_1 G_2(z)}{1 + G_1 G_2(z)} = \frac{\beta 0.00484z^{-1} + \beta 0.00468z^{-2}}{1 + (\beta 0.00484 - 0.9048)z^{-1} + (\beta 0.00468 + 0.9048)z^{-2}}$$

$$\text{(c)} \quad p(z) = z^2 + (\beta 0.00484 - 0.9048)z + (\beta 0.00468 + 0.9048)$$

Stable $\Rightarrow$ ① $a_0 > |a_2|: |\beta 0.00468 + 0.9048| < 1$

$$\Rightarrow -407.0085 < \beta < 20.3419$$

$$\begin{aligned} \text{② } p(z=1) &= 1 + \beta 0.00484 - 0.9048 + \beta 0.00468 + 0.9048 \\ &= 9.52 \times 10^{-3} \beta > 0 \Rightarrow \beta > 0 \end{aligned}$$

$$\begin{aligned} \text{③ } p(z=-1) &= 1 - \beta 0.00484 + 0.9048 + \beta 0.00468 + 0.9048 \\ &= 3.8096 - 1.6 \times 10^{-4} \beta > 0 \Rightarrow \beta < 23810 \end{aligned}$$

$$\therefore 0 < \beta < 20.3419.$$

$$\beta = 1: C(\infty) = \lim_{z \rightarrow 1} (1-z^{-1}) C(z) = 0.$$

$$\beta = 1000: C(\infty) = +\infty.$$

$$3. (a) \dot{X}(t) = A X(t), \quad \Phi = \mathcal{L}^{-1}\{[sI - A]^{-1}\}$$

$$\Rightarrow X(t) = \Phi(t) X(0)$$

$$= \begin{bmatrix} \cos t & \sin t \\ -\sin t & \cos t \end{bmatrix} \begin{bmatrix} 1 \\ 1 \end{bmatrix}$$

$$= \begin{bmatrix} \cos t + \sin t \\ \cos t - \sin t \end{bmatrix} = \begin{bmatrix} X_1(t) \\ X_2(t) \end{bmatrix}$$

$$= \mathcal{L}^{-1}\left\{ \begin{bmatrix} s & -1 \\ 1 & s \end{bmatrix}^{-1} \right\}$$

$$= \mathcal{L}^{-1}\left\{ \frac{\begin{bmatrix} s & 1 \\ -1 & s \end{bmatrix}}{s^2 + 1} \right\}$$

$$= \mathcal{L}^{-1}\left\{ \begin{bmatrix} \frac{s}{s^2+1} & \frac{1}{s^2+1} \\ -\frac{1}{s^2+1} & \frac{s}{s^2+1} \end{bmatrix} \right\}$$

$$= \begin{bmatrix} \cos t & \sin t \\ -\sin t & \cos t \end{bmatrix}$$

$$(b) \dot{X}(t) = A X(t) + B u(t),$$

$$\Phi(T) = \mathcal{L}^{-1}\{[sI - A]^{-1}\} = \mathcal{L}^{-1}\left\{ \begin{bmatrix} s & -1 \\ 1 & s+2 \end{bmatrix}^{-1} \right\} = \mathcal{L}^{-1}\left\{ \frac{\begin{bmatrix} s+2 & 1 \\ -1 & s \end{bmatrix}}{s(s+2)+1} \right\}$$

$$= \mathcal{L}^{-1}\left\{ \begin{bmatrix} \frac{s+2}{(s+1)^2} & \frac{1}{(s+1)^2} \\ \frac{-1}{(s+1)^2} & \frac{s}{(s+1)^2} \end{bmatrix} \right\} = \begin{bmatrix} (1-T)e^{-T} + \frac{1}{2}Te^{-T} & Te^{-T} \\ -Te^{-T} & (1-T)e^{-T} \end{bmatrix}$$

$$= \begin{bmatrix} e^{-T} - \frac{1}{2}e^{-T} & Te^{-T} \\ -Te^{-T} & (1-T)e^{-T} \end{bmatrix} = \begin{bmatrix} 0.1839 & 0.3679 \\ -0.3679 & 0 \end{bmatrix}$$

$$\Theta(T) = \int_0^T \Phi(\tau) B d\tau = \int_0^T \begin{bmatrix} e^{-\tau} - \frac{1}{2}e^{-\tau} & \tau e^{-\tau} \\ -\tau e^{-\tau} & (1-\tau)e^{-\tau} \end{bmatrix} d\tau$$

$$= \int_0^T \begin{bmatrix} e^{-\tau} - \frac{1}{2}\tau e^{-\tau} \\ -\tau e^{-\tau} \end{bmatrix} d\tau$$

$$X(k+1) = \Phi(T) X(k) + \Theta(T) u(k)$$

$$Y(k+1) = C X(k)$$

$$|[zI - A]| = \left| \begin{bmatrix} z - 0.1839 & -0.3679 \\ 0.3679 & z \end{bmatrix} \right| = (z - 0.1839)z + 0.3679^2$$

$$= z^2 - 0.1839z + 0.1353$$

$$\Delta = (0.1839)^2 - 4 \times 0.1353$$

$$= -0.5074$$

$$z_{1,2} = \frac{1}{2}(0.1839 \pm \sqrt{\Delta})$$

$$= 0.09195 \pm j0.7123$$

$$(c) \quad x(k+1) = Ax(k) + Bu(k) = 2x(k) + 3u(k) \quad \therefore A=2, \quad B=3$$

$$J = \frac{1}{2} \sum_{k=0}^{\infty} (x^T(k) Q x(k) + u^2(k))$$

$$= \frac{1}{2} \sum_{k=0}^{\infty} (x^2(k) + u^2(k)), \quad Q=1, \quad R=1$$

$$S = 2 \times S \times 2 + 1 - 2 \times S \times 3 \times (1 + 3 \times S \times 3)^{-1} \times 3 \times S \times 2$$

$$= 4S + 1 - 6S \times (9S + 1)^{-1} \times 6S$$

$$= 4S + 1 - 36S^2 \times (9S + 1)^{-1}$$

$$9S^2 + S = 36S^2 + 13S + 1 - 36S^2,$$

$$9S^2 - 12S - 1 = 0$$

$$\Delta = 12^2 + 4 \times 9 = 180$$

$$S_{1,2} = \frac{9}{2} (12 \pm \sqrt{180}) = \cancel{54 \pm 27\sqrt{5}}$$

$$\therefore S_1 = \cancel{54 + 27\sqrt{5}} \quad \frac{2+\sqrt{5}}{3}, \quad S_2 = \frac{2-\sqrt{5}}{3}$$

$$K = (3 \times S \times 3 + 1)^{-1} \times 3 \times S \times 2$$

$$= (9S + 1)^{-1} \times 6S$$

$$= 0.6667$$

$$\therefore u^*(k) = -0.6667x(k)$$

$$4. (a) \quad x(k+1) = Ax(k) + Bu(k) = [A - BK]x(k)$$

$$\alpha_c(z) = (z - z_1)(z - z_2) = z^2 - (z_1 + z_2)z + z_1 z_2$$

$$z_1 + z_2 = 0.5, \quad z_1 z_2 = 0.4999 \approx 0.5$$

$$\Rightarrow \alpha_c(z) = z^2 - 0.5z + 0.5$$

method 1: $k = [0 \quad 1] w_c^{-1} \alpha_c(A)$

$$w_c^{-1} = [B \quad AB]^{-1} = \begin{bmatrix} 0 & 2 \\ 1 & 3 \end{bmatrix}^{-1} = \frac{\begin{bmatrix} 3 & -2 \\ -1 & 0 \end{bmatrix}}{-2} = \begin{bmatrix} -\frac{3}{2} & 1 \\ \frac{1}{2} & 0 \end{bmatrix}$$

$$\alpha_c(A) = A^2 - 0.5A + 0.5I$$

$$= \begin{bmatrix} 1 & 2 \\ 0 & 3 \end{bmatrix} \begin{bmatrix} 1 & 2 \\ 0 & 3 \end{bmatrix} - \begin{bmatrix} \frac{1}{2} & 1 \\ 0 & \frac{3}{2} \end{bmatrix} + \begin{bmatrix} \frac{1}{2} & 0 \\ 0 & \frac{1}{2} \end{bmatrix}$$

$$= \begin{bmatrix} 1 & 8 \\ 0 & 9 \end{bmatrix} + \begin{bmatrix} 0 & -1 \\ 0 & -1 \end{bmatrix} = \begin{bmatrix} 1 & 7 \\ 0 & 8 \end{bmatrix}$$

$$\Rightarrow k = [0 \quad 1] \begin{bmatrix} -\frac{3}{2} & 1 \\ \frac{1}{2} & 0 \end{bmatrix} \begin{bmatrix} 1 & 7 \\ 0 & 8 \end{bmatrix} = \begin{bmatrix} -\frac{1}{2} & 0 \end{bmatrix} \begin{bmatrix} 1 & 7 \\ 0 & 8 \end{bmatrix} = \begin{bmatrix} \frac{1}{2} & \frac{7}{2} \end{bmatrix}$$

$$\Rightarrow u(k) = -\begin{bmatrix} \frac{1}{2} & \frac{7}{2} \end{bmatrix} x(k) = \begin{bmatrix} -\frac{1}{2} & -\frac{7}{2} \end{bmatrix} x(k)$$

method 2: $[A - BK] = \begin{bmatrix} 1 & 2 \\ 0 & 3 \end{bmatrix} - \begin{bmatrix} 0 \\ 1 \end{bmatrix} [k_1 \quad k_2] = \begin{bmatrix} 1 & 2 \\ -k_1 & 3-k_2 \end{bmatrix}$

$$|zI - [A - BK]| = \begin{vmatrix} z-1 & -2 \\ k_1 & z-3+k_2 \end{vmatrix} = (z-1)(z-3+k_2) + 2k_1$$

$$= z^2 - 3z + k_2 z - z + 3 - k_2 + 2k_1$$

$$= z^2 + (k_2 - 4)z + (3 - k_2 + 2k_1)$$

$$\alpha_c(z) = z^2 - 0.5z + 0.5$$

$$\Rightarrow k_2 - 4 = -0.5, \quad k_2 = \frac{7}{2}$$

$$3 - \frac{7}{2} + 2k_1 = 0.5, \quad k_1 = \frac{1}{2}$$

$$\Rightarrow u(k) = -\begin{bmatrix} \frac{1}{2} & \frac{7}{2} \end{bmatrix} x(k)$$

$$(b) \quad z_{1,2} = 0.1 \pm j0.1, \quad \alpha_0(z) = z^2 - 0.2z + 0.02$$

$$L_0 = \alpha_0(A) W_0^{-1} \begin{bmatrix} 0 \\ 1 \end{bmatrix},$$

$$W_0^{-1} = \begin{bmatrix} C \\ CA \end{bmatrix}^{-1} = \begin{bmatrix} 1 & 0 \\ 1 & 2 \end{bmatrix}^{-1} = \frac{\begin{bmatrix} 2 & 0 \\ -1 & 1 \end{bmatrix}}{2} = \begin{bmatrix} 1 & 0 \\ -\frac{1}{2} & \frac{1}{2} \end{bmatrix}$$

$$\alpha_0(A) = A^2 - 0.2A + 0.02I$$

$$= \begin{bmatrix} 1 & 2 \\ 0 & 3 \end{bmatrix} \begin{bmatrix} 1 & 2 \\ 0 & 3 \end{bmatrix} - \begin{bmatrix} 0.2 & 0.4 \\ 0 & 0.6 \end{bmatrix} + \begin{bmatrix} 0.02 & 0 \\ 0 & 0.02 \end{bmatrix}$$

$$= \begin{bmatrix} 1 & 8 \\ 0 & 9 \end{bmatrix} - \begin{bmatrix} 0.2 & 0.4 \\ 0 & 0.6 \end{bmatrix} + \begin{bmatrix} 0.02 & 0 \\ 0 & 0.02 \end{bmatrix} = \begin{bmatrix} 0.82 & 7.6 \\ 0 & 8.42 \end{bmatrix}$$

$$\Rightarrow L_0 = \begin{bmatrix} 0.82 & 7.6 \\ 0 & 8.42 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ -\frac{1}{2} & \frac{1}{2} \end{bmatrix} \begin{bmatrix} 0 \\ 1 \end{bmatrix} = \begin{bmatrix} -2.98 & 3.8 \\ -4.21 & 4.21 \end{bmatrix} \begin{bmatrix} 0 \\ 1 \end{bmatrix} = \begin{bmatrix} 3.8 \\ 4.21 \end{bmatrix}$$

$$(c) \quad u(k) = -K\bar{x}(k)$$

$$\bar{x}(k+1) = A\bar{x}(k) + Bu(k) + L_0(y(k) - C\bar{x}(k))$$

$$= [A - BK - L_0C] \bar{x}(k) + L_0 y(k)$$

take z-transform on both sides:

$$z\bar{X}(z) = [A - BK - L_0C] \bar{X}(z) + L_0 Y(z)$$

$$Y(z) = \frac{1}{L_0} (zI - [A - BK - L_0C]) \bar{X}(z)$$

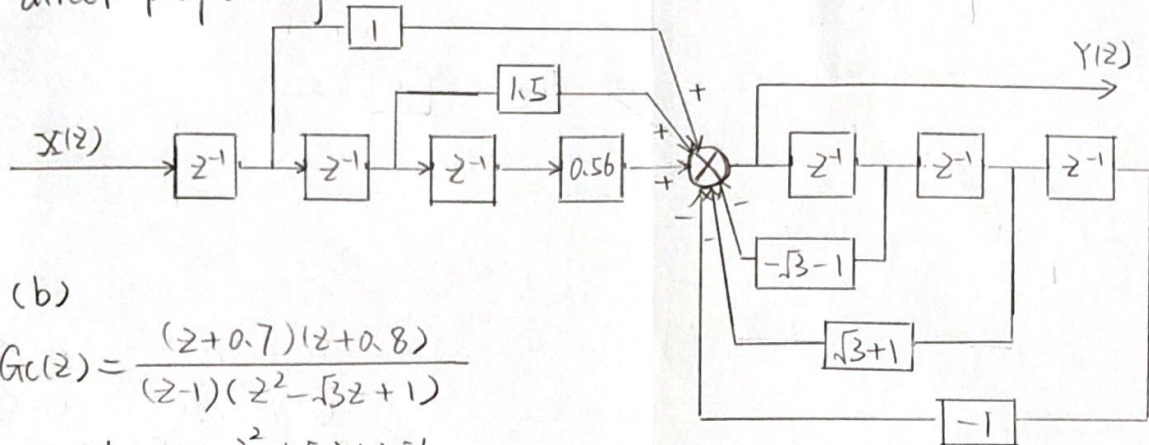
$$\Rightarrow \frac{U(z)}{Y(z)} = \frac{-K\bar{X}(z)}{Y(z)} = - \frac{KL_0}{zI - [A - BK - L_0C]}$$

$$5. (a) G_c(z) = \frac{z^2 + 1.5z + 0.56}{z^3 + (-\sqrt{3}-1)z^2 + (\sqrt{3}+1)z - 1}$$

$$\Rightarrow \frac{Y(z)}{X(z)} = \frac{z^{-1} + 1.5z^{-2} + 0.56z^{-3}}{1 + (-\sqrt{3}-1)z^{-1} + (\sqrt{3}+1)z^{-2} - z^{-3}}$$

$$\Rightarrow y(k) = (\sqrt{3}+1)y(k-1) - (\sqrt{3}+1)y(k-2) + y(k-3) + x(k-1) + 1.5x(k-2) + 0.56x(k-3)$$

direct programming:



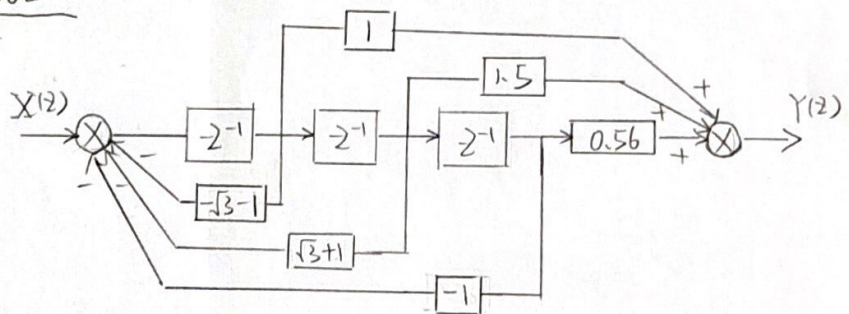
(b)

$$G_c(z) = \frac{(z+0.7)(z+0.8)}{(z-1)(z^2-\sqrt{3}z+1)}$$

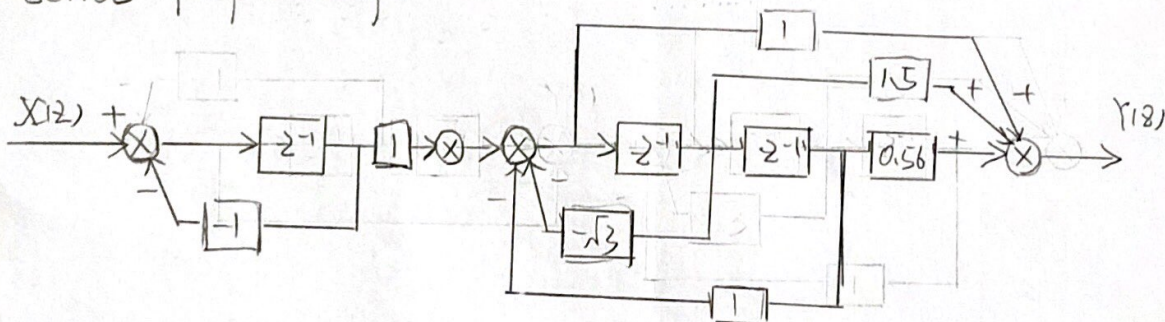
$$= \frac{1}{z-1} \times \frac{z^2 + 1.5z + 0.56}{z^2 - \sqrt{3}z + 1}$$

$$= \frac{1}{1-z^{-1}} \times \frac{1 + 1.5z^{-1} + 0.56z^{-2}}{1 - \sqrt{3}z^{-1} + z^{-2}}$$

standard programming:



series programming:



(c) comment

EE6203

NANYANG TECHNOLOGICAL UNIVERSITY
SEMESTER 1 EXAMINATION 2017-2018
EE6203 – COMPUTER CONTROL SYSTEMS

November/December 2017

Time Allowed: 3 hours

INSTRUCTIONS

1. This paper contains FIVE (5) questions and comprises EIGHT (8) pages.
2. Answer all FIVE (5) questions.
3. This is a closed-book examination.
4. All questions carry equal marks.
5. The Transform Table is included in Appendix A on pages 7 to 8.

1. (a) The Laplace Transform $X(s)$ of a signal $x(t)$ is given as

$$X(s) = \frac{1}{s^2 + 5s + 4} \cdot \frac{1}{(s+1)(s+4)} = \frac{1}{3} \frac{4-1}{(s+1)(s+4)}$$

Assume that $x(kT)$ is the sampled sequence of $x(t)$ with a sampling period T of 0.5 second. Determine the Z-transform of $x(kT)$.

$$z^2 X(z) - z^2 X(0) - z X(1) + (\alpha + 0.5) z X(z) - (\alpha + 0.5) z X(0) + 0.5 \alpha X(z) = 1 \quad (5 \text{ Marks})$$

$$(z^2 + (\alpha + 0.5)z + 0.5\alpha) X(z) - z^2 X(0) - (\alpha + 0.5)z X(0) - z X(1) = 1$$

- (b) Solve the following difference equation:

$$x(k+2) + (\alpha + 0.5)x(k+1) + 0.5\alpha x(k) = \delta(k), \text{ and } x(k) = 0 \text{ for } k < 0,$$

where α is real number and $\delta(k)$ is a unit impulse function.

(11 Marks)

- (c) Discuss the convergence of $x(k)$ when $k \rightarrow \infty$ for different values of α in part 1(b).

(4 Marks)

$$E^*(s) = R^*(s) - H^*(s) C^*(s) \\ = R^*(s) - [G_1 G_2(s)]^* H^*(s) E^*(s)$$

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2. Consider the control system shown in Figure 1.

$$E^*(s) = \frac{R^*(s)}{1 + [G_1 G_2(s)]^* H^*(s)}$$

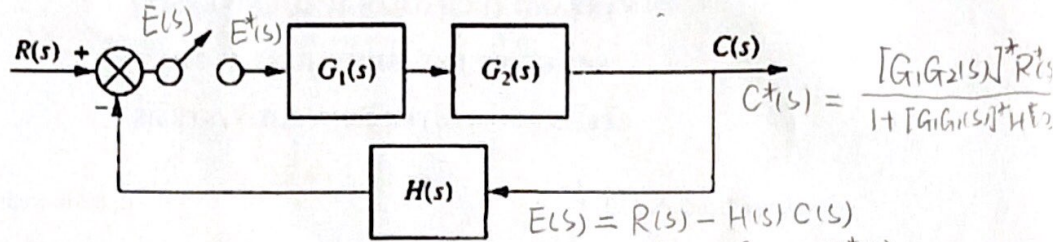


Figure 1

$$C^*(s) = \frac{[G_1 G_2(s)]^* R^*(s)}{1 + [G_1 G_2(s)]^* H^*(s)}$$

$$E(s) = R(s) - H(s) C(s) \\ C(s) = G_1(s) G_2(s) E^*(s) \\ C^*(s) = [G_1 G_2(s)]^* E^*(s)$$

- (a) Determine whether the pulse transfer function of the system exists.

(6 Marks)

- (b) Suppose that $G_1(s) = \frac{1-e^{-Ts}}{s}$, $G_2(s) = \frac{\beta}{s(s+1)}$, $H(s)=1$, the sampling period T is 0.1 second and the input is a unit-impulse function. Determine the Z-transform of the output signal.

(9 Marks)

- (c) Find the range of β in part 2(b) to ensure the stability of the system. Also, determine the output $c(kT)$ as $k \rightarrow \infty$ for $\beta = 1$ and 1000, respectively.

(5 Marks)

$$\lim_{z \rightarrow 1} (1-z^{-1}) C(z) \\ \tilde{X}(z) = \frac{1}{(z-1)^2} = \frac{z^{-2}}{(1-z^{-1})^2}$$

$$= \frac{-z^{-1}(1-z^{-1}) + z^{-1}}{(1-z^{-1})^2}$$

$$= -\frac{z^{-1}}{(1-z^{-1})} + \frac{z^{-1}}{(1-z^{-1})^2}$$

$$\downarrow \quad \downarrow \\ -(1)^{k-1} + k \cdot (1)^{k-1}$$

$$|\beta \cdot 0.00468 + 0.9048| < 1$$

$$0.00468 \beta < 0.0952$$

$$\beta < 20.3419$$

$$-0.00468 \beta < -1.9048$$

$$\beta > -407.0085$$

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3. (a) An oscillatory system can be described by the following equation

$$\begin{bmatrix} \frac{dx_1(t)}{dt} \\ \frac{dx_2(t)}{dt} \end{bmatrix} = \begin{bmatrix} 0 & 1 \\ -1 & 0 \end{bmatrix} \begin{bmatrix} x_1(t) \\ x_2(t) \end{bmatrix}$$

Determine the response $\begin{bmatrix} x_1(t) \\ x_2(t) \end{bmatrix}$ when the initial state is $\begin{bmatrix} x_1(0) \\ x_2(0) \end{bmatrix} = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$.

$x(t) = \Phi(t)x(0) + \Psi(t)$, $\Psi(t) = \mathcal{L}^{-1}[\Gamma(s)]$ (6 Marks)

- (b) A continuous-time system has a state-space representation given by $\Gamma(s) = [sI - A]^{-1}Bu(s)$

$$\begin{bmatrix} \frac{dx_1(t)}{dt} \\ \frac{dx_2(t)}{dt} \end{bmatrix} = \begin{bmatrix} 0 & 1 \\ -1 & -2 \end{bmatrix} \begin{bmatrix} x_1(t) \\ x_2(t) \end{bmatrix} + \begin{bmatrix} 1 \\ -1 \end{bmatrix} u(t)$$

$$y(t) = \begin{bmatrix} 2 & 3 \end{bmatrix} \begin{bmatrix} x_1(t) \\ x_2(t) \end{bmatrix}$$

where $x_1(t)$ and $x_2(t)$ are the states, $u(t)$ and $y(t)$ are the input and output variables, respectively. Suppose that the system is sampled with a zero-order hold at a sampling period $T = 1$ second, obtain a discretised state-space model for the system.

Is the discretised system stable?

(7 Marks)

$$(\tau^2 e^{-1})'$$

$$= 2\tau e^{-1} - \tau^2 e^{-1}$$

Note: Question No. 3 continues on page 4

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- (c) Consider a system which is described by a state-space representation

$$\mathbf{x}(k+1) = \mathbf{A}\mathbf{x}(k) + \mathbf{B}u(k)$$

The performance index for the system is given by

$$J = \frac{1}{2} \sum_{k=0}^{\infty} (\mathbf{x}^T(k) \mathbf{Q} \mathbf{x}(k) + u^2(k))$$

and the control law that minimises J is of the following form

$$u^*(k) = -\mathbf{K}\mathbf{x}(k)$$

where the optimal control law is given by

$$\mathbf{K} = (\mathbf{B}^T \mathbf{S} \mathbf{B} + 1)^{-1} \mathbf{B}^T \mathbf{S} \mathbf{A}$$

and $\mathbf{S} > 0$ solves the following

$$\mathbf{S} = \mathbf{A}^T \mathbf{S} \mathbf{A} + \mathbf{Q} - \mathbf{A}^T \mathbf{S} \mathbf{B} (1 + \mathbf{B}^T \mathbf{S} \mathbf{B})^{-1} \mathbf{B}^T \mathbf{S} \mathbf{A}$$

Determine the optimal control law for the following system and given performance index:

$$x(k+1) = 2x(k) + 3u(k)$$

$$J = \frac{1}{2} \sum_{k=0}^{\infty} (x^2(k) + u^2(k))$$

(7 Marks)

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4. A discrete-time system has a state-space representation given by

$$\begin{bmatrix} x_1(k+1) \\ x_2(k+1) \end{bmatrix} = \mathbf{A} \begin{bmatrix} x_1(k) \\ x_2(k) \end{bmatrix} + \mathbf{B} u(k)$$

$$y(k) = \mathbf{C} \begin{bmatrix} x_1(k) \\ x_2(k) \end{bmatrix}$$

where

$$\mathbf{A} = \begin{bmatrix} 1 & 2 \\ 0 & 3 \end{bmatrix}, \quad \mathbf{B} = \begin{bmatrix} 0 \\ 1 \end{bmatrix}, \quad \mathbf{C} = [1 \quad 0]$$

$\begin{bmatrix} x_1(k) \\ x_2(k) \end{bmatrix}$, $u(k)$ and $y(k)$ are the states, input and output variables, respectively.

- (a) Determine the following control law

$$u(k) = -\mathbf{K}\mathbf{x}(k)$$

such that the closed-loop poles are at $z_{1,2} = 0.25 \pm j0.6614$.

(7 Marks)

- (b) Design a full-order prediction estimator of the following form

$$\bar{\mathbf{x}}(k+1) = \mathbf{A}\bar{\mathbf{x}}(k) + \mathbf{B}u(k) + \mathbf{L}_e(y(k) - \mathbf{C}\bar{\mathbf{x}}(k))$$

such that the desired estimator poles are at $z_{1,2} = 0.1 \pm j0.1$.

(7 Marks)

- (c) The true state $\mathbf{x}(k)$ in the control law in part 4(a) is replaced with the estimated state $\bar{\mathbf{x}}(k)$ from the estimator in part 4(b). Obtain the combined controller-estimator

transfer function $\frac{U(z)}{Y(z)}$ in terms of $\mathbf{A}$, $\mathbf{B}$, $\mathbf{C}$, $\mathbf{L}_e$ and $\mathbf{K}$, where applicable.

(6 Marks)

EE6203

5. (a) A digital controller is designed for a particular industrial process based on certain specifications. Its pulse transfer function is given as follows:

$$G_c(z) = \frac{U(z)}{E(z)} = \frac{(z+0.7)(z+0.8)}{(z-1)(z^2 - \sqrt{3}z + 1)}$$

Draw a block diagram representation if it is desired to implement the controller using the standard programming approach.

(10 Marks)

- (b) Implement the controller $G_c(z)$ in part 5(a) with the series programming approach and show the relevant block diagrams.

(8 Marks)

- (c) Comment on the two implementation approaches in parts 5(a) and 5(b).

(2 Marks)